

Predicting Economic Damage of Natural Disasters

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Invalid Date

Data processing

Introduction

Background:

As global temperatures rise, climate-related disasters have increased in both frequency and severity over the past several decades ([World Wild Life](#)). Extreme weather events such as floods, droughts, hurricanes, etc. impose a huge human and real economic costs across the world. It is imperative that we understand the determinants of disaster severity (in terms of economic damages) for better policy planning and mitigative strategies.

The Emergency Events Database (EM-DAT), maintained by the Centre for Research on the Epidemiology of Disasters (CRED), keeps an internationally standardized data on natural disasters worldwide. The information includes disaster type, location, affected population, deaths, injuries, and economic damages.

Research Question:

How do different disaster characteristics (such as geographic location, time of year, type, and total casualties incurred) predict the total economic damage caused by natural disasters in USD?

- Specifically, we examine whether disaster type, location, timing, duration, and measures of human impact (deaths, injuries, and affected population) are associated with higher reported economic losses.

Motivation:

We were first drawn to this research question because we were intrigued by this data; we were amazed by the granularity and the surprising breadth of information in this data. However, we realized that doing research on this matter could have real, practical uses.

First, identifying predictors of large economic damage is critical for both public and private institutions. Governments and international organizations allocate substantial resources towards mitigation and recovery efforts, yet the determinants of high-cost disasters vary across event types and regions. Same applies for insurance companies and hedge fund traders that try modeling key determinants relevant to a disaster for their comparative advantage against peers in the market.

By analyzing global disaster data, this project aims to quantify which observable characteristics are most strongly associated with economic loss. We hope that these results may provide insight into which types of disasters and impact measures are most economically destructive.

Hypothesis Reasoning:

Based on economic reasoning and relevant research, we expect that...

1. Disasters with higher number of affected individuals (absolute number, number of deaths and injuries) will have a positive relationship with economic damage.
2. Certain disaster types will incur systematically higher economic damage than that of others.
3. The level of damage dollars will differ based on countries because of their respective infrastructure that exists to mitigate damage.

Data Description:

This data was obtained from the Emergency Events Database, otherwise known as [EM-DAT](#), which is maintained by the Centre for Research on the Epidemiology of Disasters (CRED). The goal of the database is to compile standardized information on natural disasters that occur around the world. According to the information on metadata, an event is included in EM-DAT if one or more of the following criteria is met:

- 10+ people killed
- 100+ people affected
- Declaration of a state of emergency
- Solicitation of international assistance

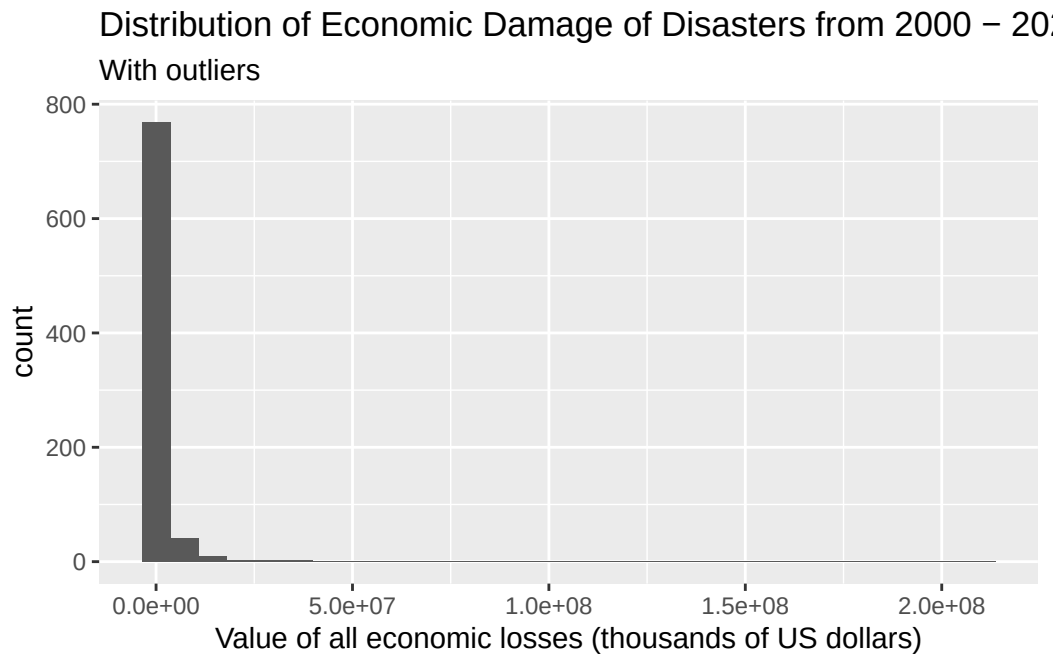
The observations describe different disasters that occurred around the world. Of each disaster, data on the number of people injured, affected, homeless, or killed, the total economic damages, type of disaster, exact location, start date and end date of the disaster were collected.

The data dictionary can be found [here](#).

Limitations and Concerns

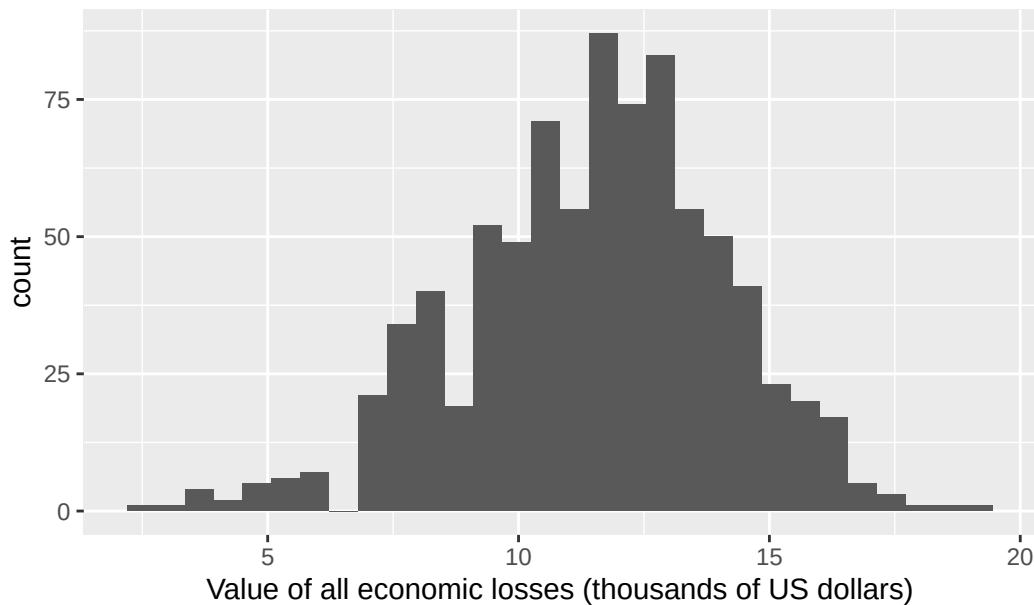
There are a bunch of outliers in our data set in terms of the total economic damage, duration in days, and many other variables. This motivated us to apply the logarithmic function to those variables. These outliers, however, should still be kept in mind.

Exploratory Data Analysis



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
12	20000	109576	1519107	559750	210000000

Distribution of Economic Damage of Disasters from 2000 – 202



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2.485	9.903	11.604	11.493	13.235	19.163

```
[1] 2.606016
```

The distribution of the unchanged response variable is right-skewed and unimodal. The center is best represented by the median, which is \$109,576,000, and the spread is best represented with the IQR, which is \$539,750,000 (559,750,000 - 20,000,000). There are 123 (828-705) outliers present in the upper end of our distribution.

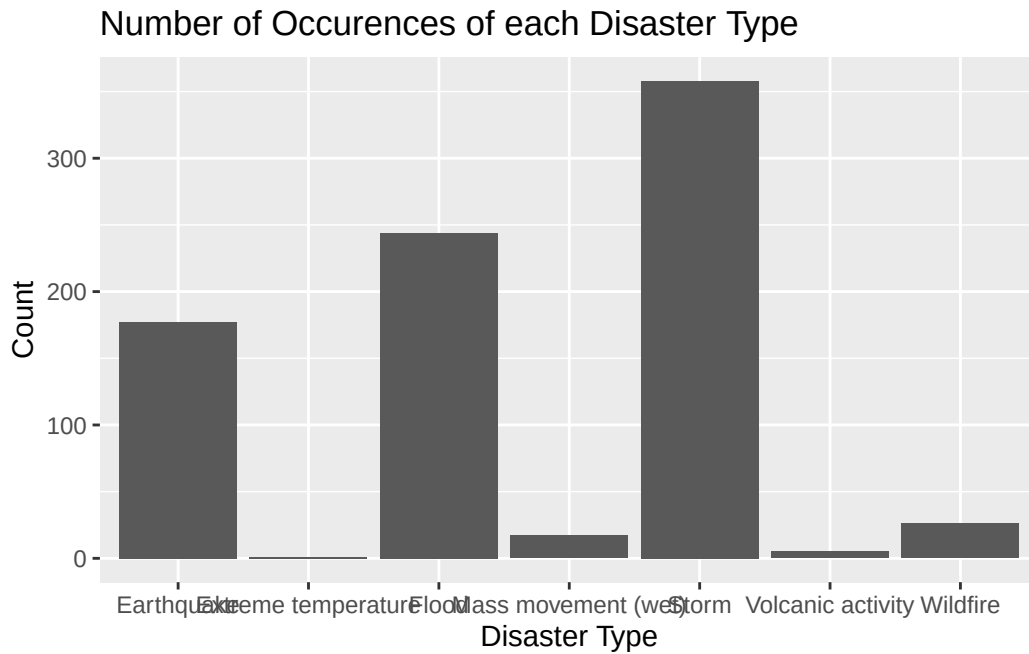
The distribution of the logged response variable is unimodal and symmetric. The center is best represented by the mean, which is \$98,027,173.52 ($e^{11.493} * 1000$), and the spread is best represented with the standard deviation, which is \$13544.98 ($e^{2.606016} * 1000$). There are outliers present in our distribution. **We will be working with the logged response variable**, log of total economic damage in thousands of USD, because without the log, the outliers skew the data heavily.

Predictors of interests:

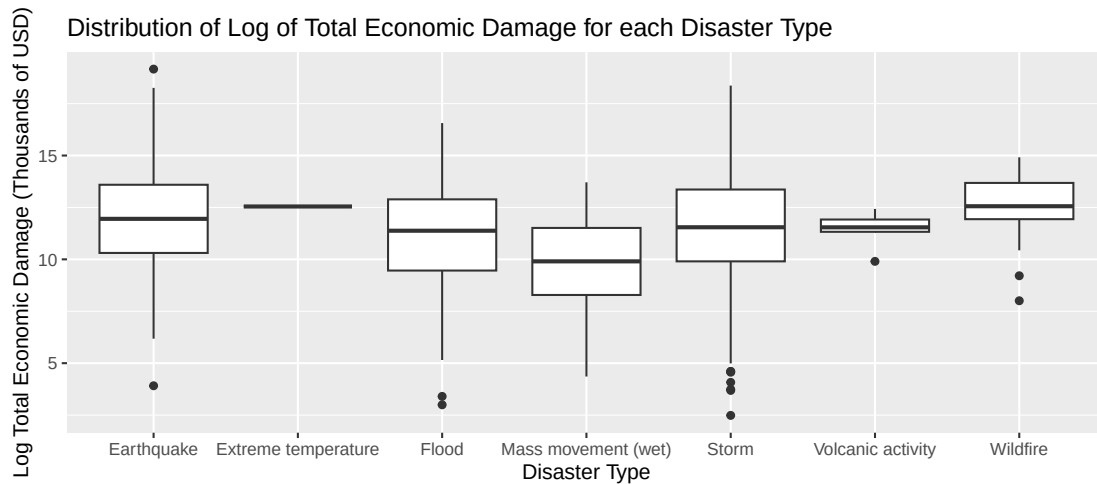
1. disaster_type - the type of disaster (categorical)

```
# A tibble: 7 x 1
  disaster_type
```

```
<chr>
1 Flood
2 Storm
3 Earthquake
4 Mass movement (wet)
5 Wildfire
6 Volcanic activity
7 Extreme temperature
```



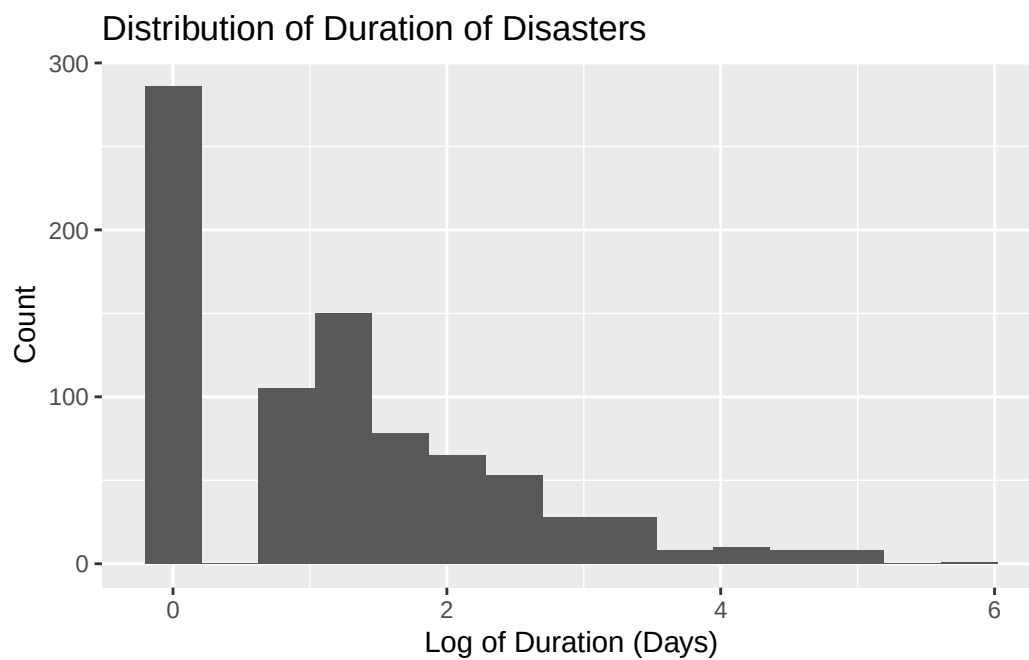
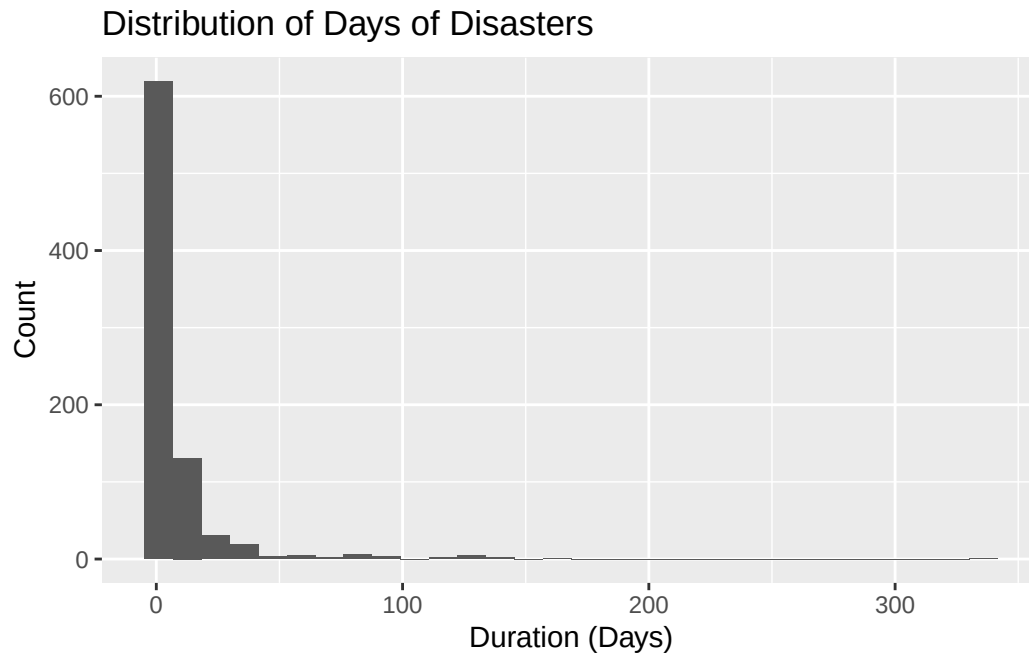
Storms, floods, and earthquakes are the most common disaster types, composing most of the data. Storms are the most common. Extreme temperature, wet mass movements, volcanic activity, and wildfires are the least common disaster types, with extreme temperature being the least common.



```
# A tibble: 7 x 3
  disaster_type      median    IQR
  <chr>            <dbl> <dbl>
1 Earthquake       12.0  3.28
2 Extreme temperature 12.5  0
3 Flood            11.4  3.43
4 Mass movement (wet)  9.90 3.23
5 Storm            11.5  3.46
6 Volcanic activity  11.5  0.592
7 Wildfire         12.6  1.75
```

Wildfires have the highest median log of total economic damage (in thousands of dollars), which is 12.558857, and thus, normal total economic damage, which would be 284604.84 thousands of dollars ($e^{12.558857}$). The median of earthquakes is slightly lower, and then the median of volcanic activities, floods, and storms, which all have similar values, slightly below the median of earthquakes. Wet mass movements have the lowest median (log of) total economic damage. Storms, floods, and earthquakes have the highest variability in the (log of) total economic damage.

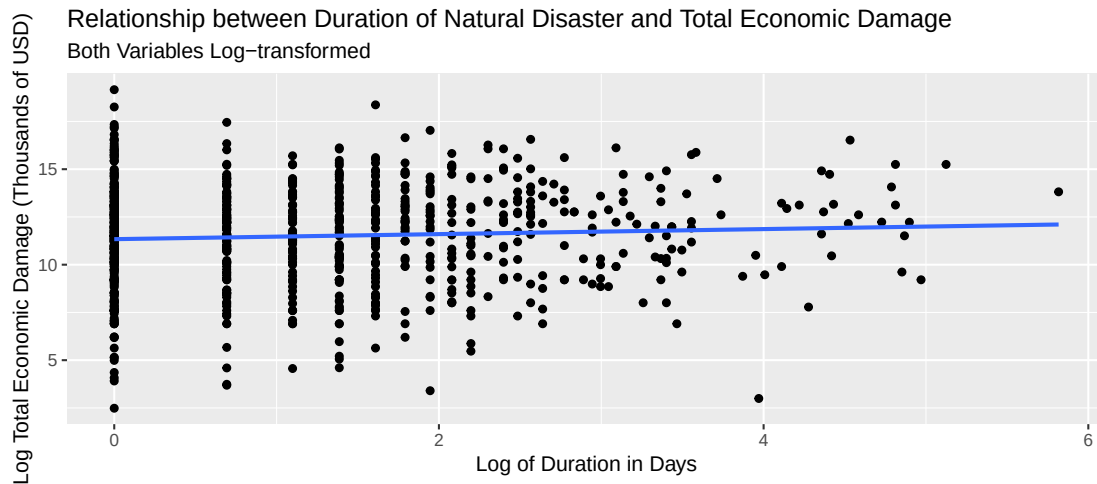
2. duration - duration in days of the disaster (quantitative)



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	1.099	1.187	1.946	5.817

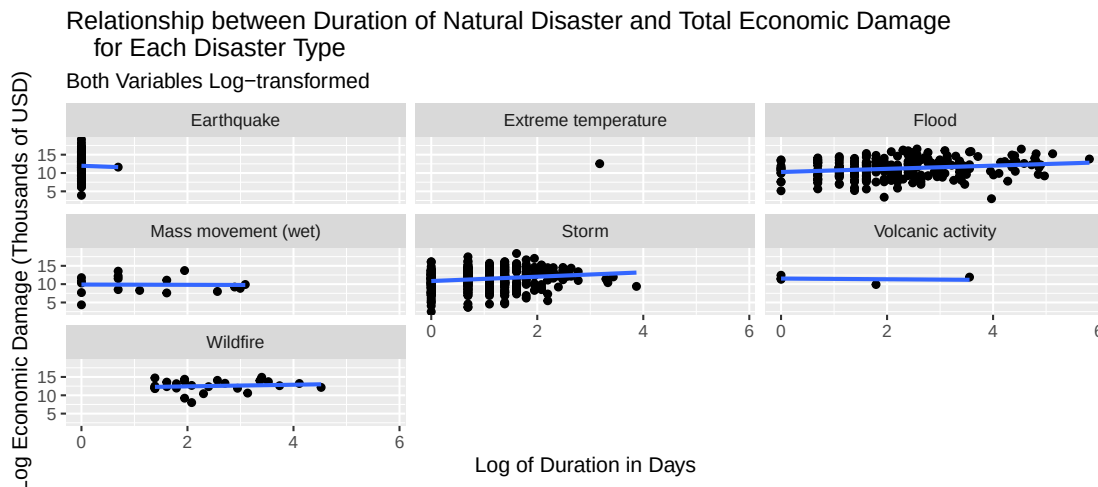
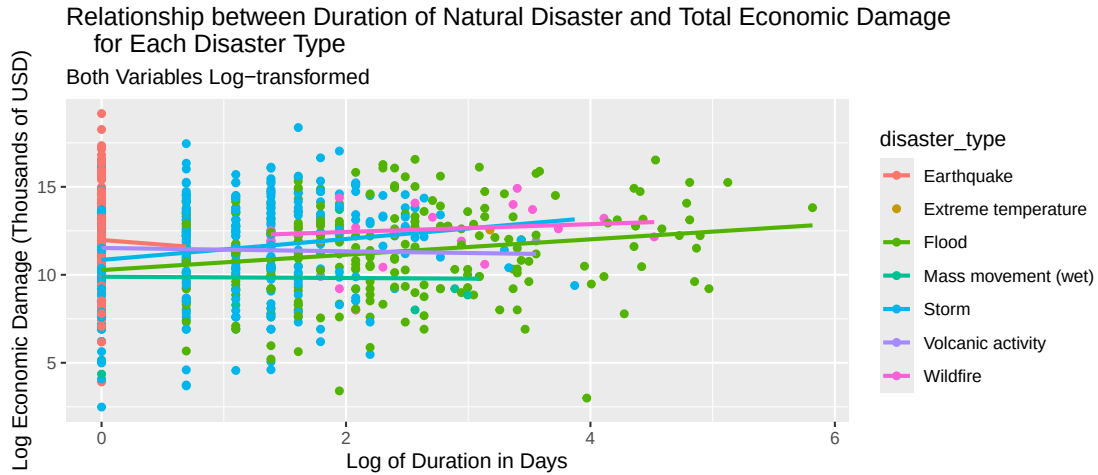
Because of the extreme outliers and extreme skew in the histogram of the duration (in

days) of each disaster, we decided to explore the log of the duration of the disaster. The histogram of the log of the duration has right skew and is bimodal. This makes sense because a lot of the disasters only last 1 day long due to their inherent nature, such as earthquakes. The median is 1.099 log duration in days, and the IQR is 1.946 log duration in days. There are outliers in the upper end of our distribution.



```
# A tibble: 1 x 1
  r_squared
    <dbl>
1 0.00360
```

There seems to be a weak positive linear relationship between the log of duration in days with log of total economic damage (in thousands of dollars). The r-squared value is 0.003596655. There is a lot of variability in the response variable when the log of duration in days is low, but the variability decreases as the x value increases. There are outliers present throughout the graph.



After visualizing the potential interaction effect between log duration of days and type of disaster, it is evident that an increase in the log duration in days has a stronger effect for storms and floods, compared to wildfires, volcanic activities, and wet mass movements, which is shown by the slightly steeper positive slope. This makes logical sense as well. We will consider putting this interaction effect into our model because of this visualization.

3. total_deaths - Total fatalities (deceased and missing combined) (quantitative)
4. no_injured - Number of people with physical injuries, trauma, or illness requiring immediate medical assistance due to the disaster (quantitative)
5. no_affected - Number of people requiring immediate assistance due to the disaster (quantitative)

! Important

Before you submit, make sure your code chunks are turned off with `echo: false` and there are no warnings or messages with `warning: false` and `message: false` in the YAML.